

Do two of these in Mathematica and the other two in Python.

1. Suppose, for an unknown relationship, $y(x)$, we have the data points

$$x = \{1.0, 2.5, 3.5, 7.0, 11.2, 15.1\}$$

$$y = \{2.6, 4.8, 5.1, 4.9, 9.6, 12.3\}$$

- (a) Using linear interpolation, determine $y(3.0)$ and $y(9.0)$.
 - (b) Construct a cubic spline fit and determine $y(3.0)$ and $y(9.0)$. Compare to your result from (a)
2. A weak x-ray emission peak is superimposed on top of a Bremsstrahlung background. The intensity of emission is measured as a function of wavelength:

$$\lambda(\text{\AA}) = \{2.5, 2.6, 2.7, 2.8, 2.9, 3.0, 3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 4.0, 4.1, 4.2, \\ 4.3, 4.4, 4.5, 4.6, 4.7, 4.8, 4.9, 5.0, 5.1, 5.2, 5.3, 5.4, 5.5, 5.6, 5.7, 5.8, 5.9, 6.0\}$$

$$I(\text{arbitrary units}) = \{0.00, 0.89, 1.65, 2.31, 2.91, 3.52, 4.18, 4.92, 5.60, 6.02, 6.09, 5.91, \\ 5.70, 5.59, 5.58, 5.64, 5.71, 5.78, 5.84, 5.89, 5.93, 5.95, 5.98, 5.99, \\ 6.00, 6.01, 6.00, 5.99, 5.98, 5.97, 5.95, 5.93, 5.91, 5.88, 5.86, 5.83\}$$

- (a) Plot the data, and from the plot estimate the wavelength of the emission peak.
 - (b) Create a spline fit for the background.
 - (c) Subtract the background to isolate the emission peak.
 - i. Fit this peak to a Gaussian, and estimate the peak.
 - ii. Compare this peak to your estimate from the original data plot.
3. The position of a cart sliding down an air track is recorded as a function of time.

$$t(\text{s}) = \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, \\ 1.9, 2.0, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3.0, 3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7, \\ 3.8, 3.9, 4.0, 4.1, 4.2, 4.3, 4.4, 4.5, 4.6, 4.7, 4.8, 4.9, 5.\}$$

$$x(\text{cm}) = \{6.29, 6.52, 6.58, 6.79, 6.9, 6.89, 7.05, 7.00, 7.18, 7.36, 7.58, 7.72, 7.78, 7.95, 7.91, \\ 8.18, 8.34, 8.57, 8.67, 8.78, 9.07, 9.06, 9.48, 9.58, 9.63, 9.9, 10.23, 10.32, 10.44, \\ 10.82, 10.95, 11.2, 11.39, 11.63, 11.98, 12.15, 12.49, 12.62, 12.93, 13.22, 13.36, \\ 13.57, 13.85, 14.26, 14.42, 14.68, 15.13, 15.48, 15.6, 15.9, 16.22\}$$

- (a) Numerically differentiate this data to find and plot $v(t)$.
 - i. Fit this to a line.

- ii. Try smoothing the data by averaging over 3 points or 5 points. Fit the smoothed graph to a line and compare. Do this by smoothing the original data, and then by smoothing the differentiated data.
- (b) Calculate the second derivative to find $a(t)$.
 - i. Plot your results and comment.
 - ii. Calculate the mean and standard deviation.
 - iii. Does it matter whether you use the original unsmoothed data, or the smoothed data?
- (c) Fit the original data to a quadratic.
 - i. Differentiate this fit function to find $v(t)$, and compare to your previous results.
 - ii. Differentiate to find $a(t)$, and compare to your previous results.
- 4. A spring has one end fixed, while the other end is pulled with a variable force, F . The amount the spring stretches, x , is measured:

$$x(\text{cm}) = \{5.0, 10.0, 15.0, 20.0, 25.0, 30.0, 35.0, 40.0\}$$

$$F(\text{N}) = \{1.9, 3.7, 5.6, 7.3, 9.2, 11.0, 12.9, 14.6\}$$

- (a) Construct a linear fit to find the spring constant, k .
- (b) Integrate $\int Fdx$, using data points:
 - i. Use a rectangular approximation.
 - ii. Use the trapezoidal approximation.
- (c) Compare your results to $\frac{1}{2}kx^2$.
- (d) Integrate your fit function, and compare to your results above.